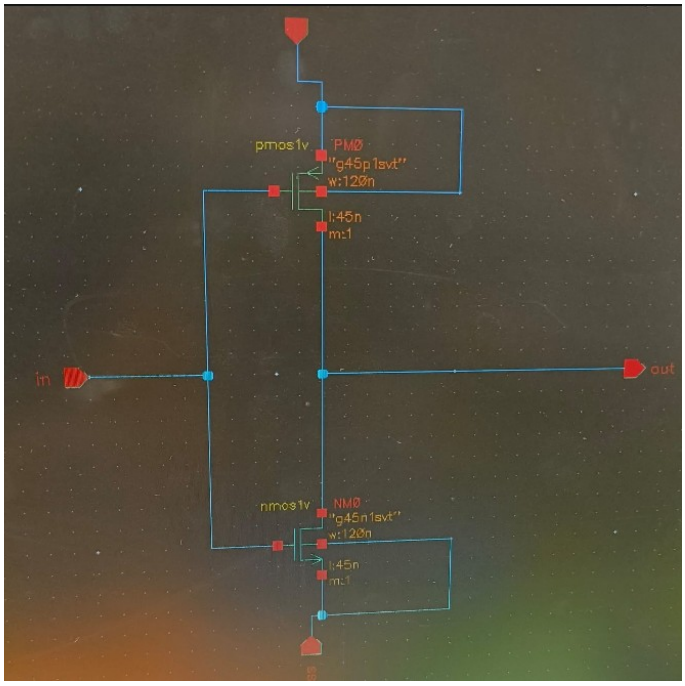
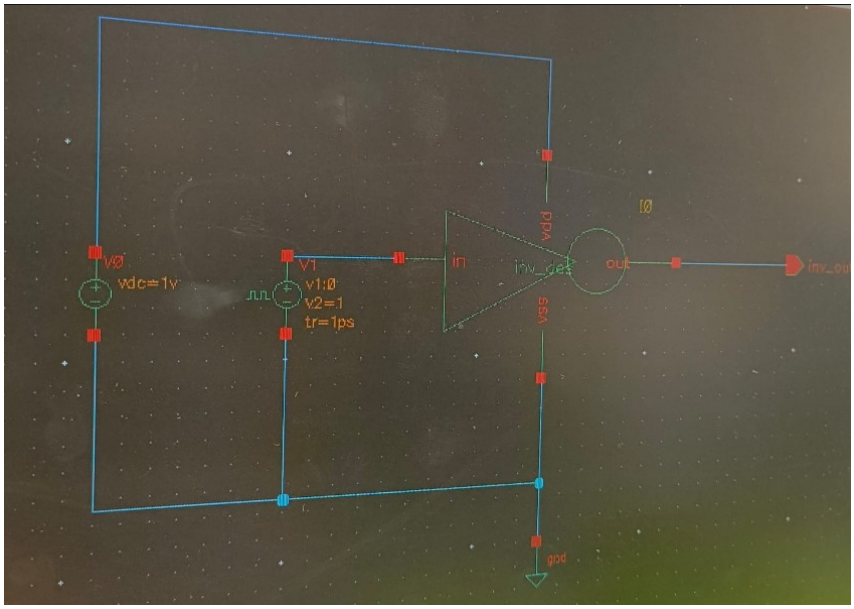
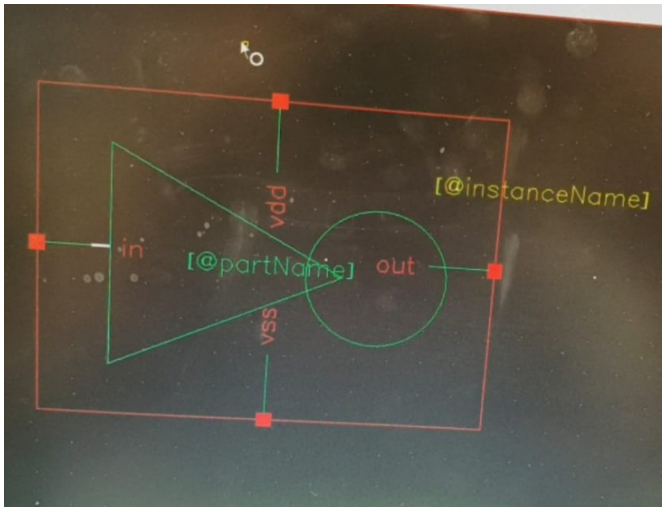
EXPERIMENT 2

Simulation-Based Analysis of P-Well CMOS Formation Process Using Cadence Design Environment

Circuit diagram:



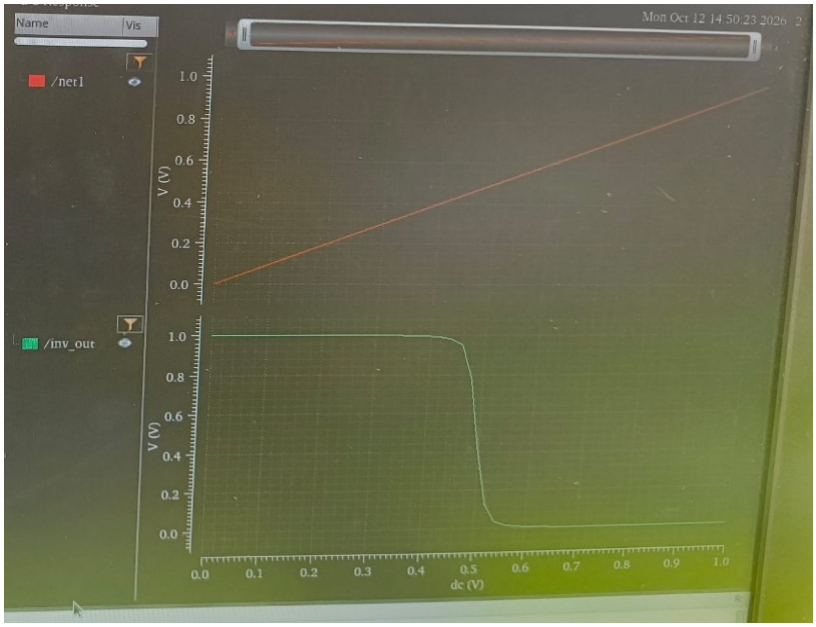




Output Parameter Table:

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	Fabrication sequence	Standard P-well CMOS flow	CMOS P-well fabrication sequence completed	0% deviation
2	P-well depth	As specified by process	2 µm	0% deviation
3	Oxide thickness	4 nm	4 nm	0% deviation
4	P-well lateral dimension	As per design	2 µm	0% deviation
5	Junction profile	Properly formed	P-well region formed in an n-type silicon substrate Record observation	0% deviation
6	Layer structure	Correct CMOS layers	N-type substrate, oxide, mask and P-well regions identified	0% deviation
7	Doped-region concentration	As per implant specification	P-well dimensions verified against the design	0% deviation

Output:


Name
Vis
Mon Oct 12 14:50:23 2026
/net1
/inv_out
V (V)
V (V)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
dc (V)